III. Methodology

In Fig 1, chronological splitting and harmful hardening are used to secure the data. While the centralized ones provide baselines, the federated architectures use calibrated output for selective alert generation and XAI verification

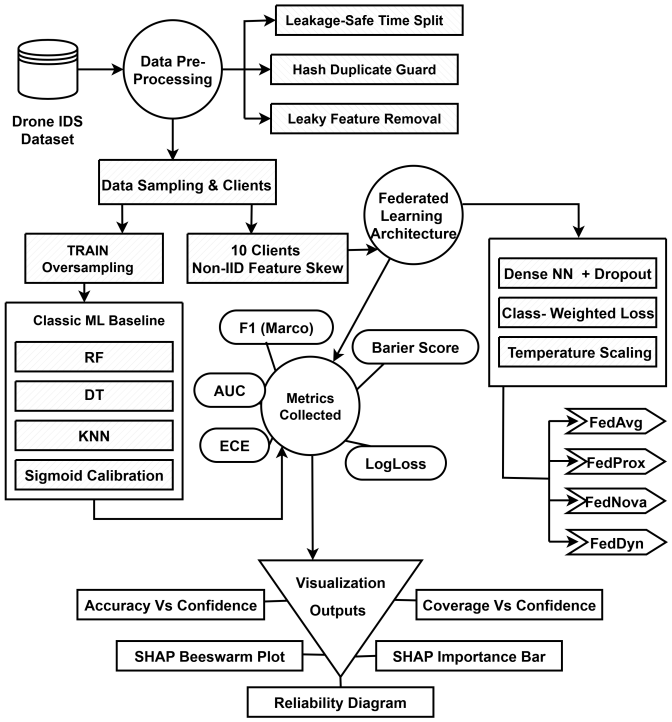

Fig. 1. Model development pipeline

A. Dataset Description and Binary Formulation

The ISOT Drone Intrusion dataset provides 301,555 telemetry records with binary labels [**?**]. This binary formulation supports centralized evaluation and realistic federated detection under feature skew.

B. Leakage-Safe Data Engineering Pipeline

1) Chronological Holdout Splits: Chronological splitting reflects real deployment by training on past telemetry to evaluate future observations.

2) Cross-Split Duplicate Hashing: A duplicate guard secures partitions against structural memorization by processing train, validation, and test splits.

3) Feature Hardening: Feature hardening is used to remove brittle and overly predictive signals. Constant columns are discarded, and a univariate ROC-AUC screening step is applied to numeric features.

4) Train-Only Oversampling: Class imbalance in the centralized baselines is addressed using oversampling applied only to the training set.

5) Preprocessing Summary and Reproducibility: The complete pipeline logs experimental parameters while applying a single hardened feature space consistently across evaluation tracks.

C. Intrusion Detection Models and Federated Learning Algorithms

1) Centralized Baselines: Centralized baselines are trained to provide a reference upper bound for detection performance. In this work, Decision Tree (DT), Random Forest (RF), and K-Nearest Neighbors (KNN) are considered as classical machine learning baselines.

2) Threshold Optimization via Youden's J Statistic: Decision thresholds are tuned on the validation set using Youden's J statistic, which balances sensitivity and specificity:

$$J(t) = \mathrm{TPR}(t) + \mathrm{TNR}(t) - 1, \tag{1}$$

3) Non-IID Client Construction: To simulate severe feature heterogeneity, the training data are partitioned into ten federated clients using shards sorted according to the continuous `Srate` attribute. This procedure creates clients with shifted marginal feature distributions, thereby inducing non-IID feature skew.

4) Local Model Architecture: A compact neural network is used as the local model in the federated setting to support efficient deployment under drone resource constraints. The architecture is composed of dense layers with dropout regularization. Local training minimizes class-weighted binary cross-entropy using a fixed learning rate.

5) Federated Aggregation Algorithms: Four federated aggregation strategies are evaluated to address client heterogeneity.

a) FedAvg: The server aggregates local model parameters across K clients utilizing sample size n_k.

$$w^{(r+1)} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{(r+1)}, \qquad n = \sum_{k=1}^{K} n_k, \tag{2}$$

b) FedProx: FedProx stabilizes local training using proximal regularization, where μ controls penalty strength

$$\min_{w} \; F_k(w) + \frac{\mu}{2}\|w - w^{(r)}\|^2, \tag{3}$$

c) FedNova: FedNova normalizes updates before global aggregation, where τ_k denotes effective local steps, written as

$$\Delta_k = w_k^{(r+1)} - w^{(r)}, \qquad \hat{g}_k = \frac{\Delta_k}{\tau_k}, \tag{4}$$

$$w^{(r+1)} = w^{(r)} + \tau \sum_{k} \frac{n_k}{n} \hat{g}_k. \tag{5}$$

d) FedDyn: FedDyn incorporates dynamic regularization through the client-specific state variable h_k before updating.

$$\min_{w} \; F_k(w) - \langle h_k, w \rangle + \frac{\alpha}{2}\|w - w^{(r)}\|^2, \tag{6}$$

$$h_k \leftarrow h_k + \alpha \left(w_k^{(r+1)} - w^{(r)} \right). \tag{7}$$

6) Federated Training Protocol: Federated training includes eight communication rounds with two local epochs using server-broadcasted models. After clients perform local optimization, the server aggregates parameters, evaluating the final model using calibrated probability outputs.

D. Probability Calibration

Calibration improves confidence trustworthiness via validation-only processing, securing scores without retraining weights.

1) Classical Model Calibration: For classical centralized models such as Random Forest, post-hoc sigmoid calibration is applied to the validation predictions. This preserves score ordering while adjusting confidence sharpness. Temperature scaling is performed as outlined in Algorithm 1.

2) Temperature Scaling for Federated Learning: Temperature scaling functions as a necessary post-processing calibration technique for federated neural models.

Temperature scaling procedures follows Algorithm 1:

Algorithm 1 Temperature Scaling for Post-Hoc Calibration

1) Generate candidate temperature values $T > 0$.
2) Convert raw logits into probabilities.
3) Compute log-loss for each candidate.
4) Select T^* minimizing validation loss.
5) Apply T^* to test predictions.

E. Evaluation Metrics

The evaluation framework measures both discriminative performance and probabilistic reliability.

Accuracy: This metric represents every correct prediction.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{8}$$

Precision: It represents the exact accuracy of positive alerts.

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{9}$$

Recall: It measures the ability to identify all actual positive.

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{10}$$

Specificity: The true negative ratio identifies actual negative classes.

$$\text{Specificity} = \frac{TN}{TN + FP}. \tag{11}$$

F1-score: It balances precision and recall despite imbalances.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{12}$$

Macro-F1 (binary): It reliably balances precision and recall.

$$F1_{\text{macro}} = \frac{F1_{\text{Attack}} + F1_{\text{Normal}}}{2}. \tag{13}$$

ROC-AUC: It determines class distinction using integrals.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}). \tag{14}$$

LogLoss: It penalizes confident errors using cross-entropy.

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]. \tag{15}$$

Brier Score: It measures squared probability differences.

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2. \tag{16}$$

Expected Calibration Error (ECE): It averages gaps between confidence and empirical accuracy where empirical accuracy over M bins:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \text{acc}(B_m) - \text{conf}(B_m) \right|. \tag{17}$$

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(\hat{y}_i = y_i), \tag{18}$$

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} p_i. \tag{19}$$

Selective alerting follows procedures in Algorithm 2:

Algorithm 2 Selective Alerting with Confidence-Based Deferral

1) Extract confidence from predicted probabilities
2) Filter samples using confidence thresholds
3) Classify samples using tuned thresholds
4) Evaluate accuracy and overall coverage
5) Plot accuracy and coverage tradeoff curves

F. Explainability Method

Explicit Features After Feature Engineering Explainability analysis Inspects learned logic hardening , checking robust markers without making any kind of indicator alteration to predictions.

1) SHAP for Global Explanations: SHAP for Global Explanations: SHAP is based on global was a beeswarm plot to compare centralized and federated telemetry plots. prioritization. This analysis shows whether defensive logic is dependent or not upon brittle identifiers, structural anomalies.

IV. Results and Discussion

A. Experimental Setup

Experiments were executed in Google Colab using Python and standard machine learning libraries. Training was done with 8 rounds across the globe with 2 local epochs.

B. Comparative Performance Evaluation

1) Data Summary and Leakage-Safe Splits: Table I illustrates eliminating 651 duplicates from the 301,555 records secured 211,088 unbiased training instances.

TABLE I
Dataset Description and Leakage-Safe Splits

Split	Total Samples	Normal (0)	Attack (1)	Robust Features
Global	301,555	89,648	211,907	63
Train	211,088	43,324	167,764	49
Validation	29,505	6,189	23,316	49
Test	60,311	40,135	20,176	49

2) Centralized Results (DT / RF / KNN): Table II shows classic models using training oversampling and Youden's J threshold optimization. The primary Macro F1 reached 0.9896, whereas KNN had hit 0.9801 and trees resulted in 0.9589.

TABLE II
CENTRALIZED BASELINE PERFORMANCE ON TEST

Metric	RF	KNN	DT
Accuracy	0.9907148	0.9842981	0.9715806
Precision	0.9802384	0.9464254	0.9270450
Recall	0.9896433	0.9997522	0.9932096
Macro F1	0.9896433	0.9801165	0.9589874
ROC-AUC	0.9998569	0.9997453	0.9769590
LogLoss	0.0391065	0.0824983	0.4580664
Brier	0.0068225	0.0093469	0.0284188
ECE	0.0163936	0.0051908	0.0284188
Tuned Thr	0.47	0.27	0.05

Table III presents the post-hoc sigmoid calibration mapping. While accuracy remained 0.9906, dropping ECE to 0.0129 improved actual probability quality.

TABLE III
CALIBRATED RANDOM FOREST PERFORMANCE ON TEST

Metric	RF (Calibrated)
Accuracy	0.9906319
Precision	0.9801189
Recall	0.9976207
Macro F1	0.9895515
ROC-AUC	0.9998585
LogLoss	0.0385591
Brier	0.0068268
ECE	0.0129036
Tuned Thr	0.45

3) Federated Results under Non-IID Skew (FedAvg / FedProx / FedNova / FedDyn): Ten state-based clients with equal numbers of shards trained 8 federated rounds with 2 Pocets.

Table IV illustrates that all the evaluated federated variants had achieved near-perfect recall for malicious attack traffic. FedAvg and FedNova reduced false positive, and was able to achieve better Macro-F1 and specificity scores overall.

TABLE IV
CONFUSION MATRICES ON TEST (ATTACK IS POSITIVE)

Model	TN	FP	FN	TP
FedAvg	39,296	839	5	20,171
FedNova	39,266	869	1	20,175
FedProx	38,253	1,882	2	20,174
FedDyn	38,158	1,977	1	20,175

Table V shows FedAvg got 0.9860 accuracy and 0.9844 Macro F1 under severe stress. While FedNova performed about the same, with lower alternative scores indicating excessive regularization kills protocol adaptation.

Fig. 2 plots the validation scores of Macro-F1 by communication rounds. FedAvg set themselves on a dominant trajectory from the outset, whereas FedDyn displayed signs of instability before getting better.

TABLE V
FEDERATED PERFORMANCE UNDER SRATE FEATURE-SKEW

Metric	FedAvg	FedNova	FedProx	FedDyn
Accuracy	0.9860059	0.9855748	0.9687619	0.9672033
Precision	0.9600666	0.9587056	0.9146717	0.9107530
Recall	0.9997522	0.9999504	0.9999009	0.9999504
Specificity	0.9790956	0.9783481	0.9531083	0.9507412
Macro F1	0.9844413	0.9839684	0.9656778	0.9640030
ROC-AUC	0.9996446	0.9997361	0.9996082	0.9986859
LogLoss	0.0169464	0.0184572	0.0304564	0.0753883
Brier	0.0030107	0.0028554	0.0036457	0.0102416
ECE	0.0026675	0.0031123	0.0165939	0.0495116
Thr	0.05	0.05	0.06	0.15
Temp T	2.0	2.0	2.0	1.5

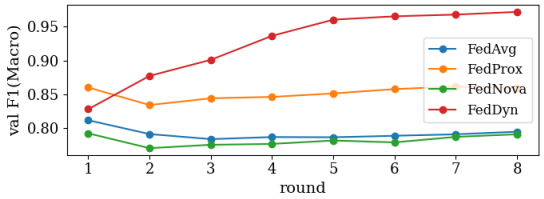

Fig. 2. Validation Macro-F1 across FL communication rounds.

4) Calibration Impact: Fig. 3 shows the work of calibration to bring the empirical hit rates to the ideal diagonal. Temperature scaling reduced FedAvg ECE to 0.0026675, which was comparable to confidence with likelihoods of threats.

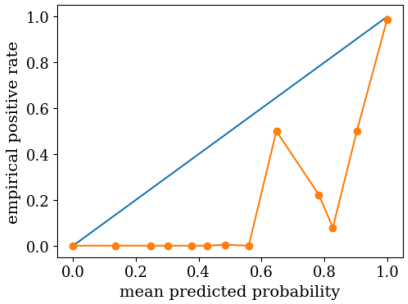

Fig. 3. Reliability diagram of centralized RF (calibration quality).

5) Selective Alerting Results: Table VI shows deferring uncertain predictions using calibrated confidence significantly improves FedAvg's precision. By increasing the threshold to 0.98 we get 0.9980 accuracy with 98.0% coverage.

TABLE VI
SELECTIVE ALERTING TRADEOFF FOR FEDAVG

Min Conf.	Acc. on Kept	Coverage
0.50	0.9860	1.000
0.98	0.9980	0.980

Fig. 4 and 5 show the tradeoff between confidence threshold and accuracy of the predictions, and retained coverage. As low confidence samples are deferred, the accuracy on the remain ing flows is increased, and the proportion of processed traffic naturally decreases.

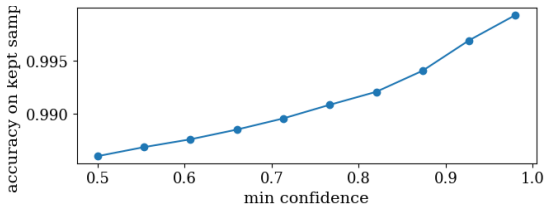

Fig. 4. Selective alerting tradeoff: accuracy vs. confidence threshold.

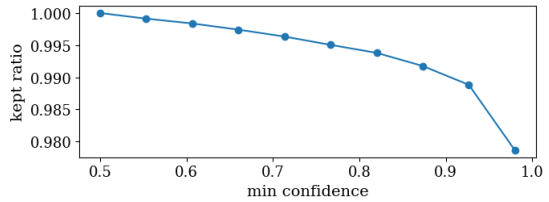

Fig. 5. Selective alerting tradeoff: coverage vs. confidence threshold.

6) XAI Result Summary: Fig. 6 presents the SHAP beeswarm summary from the best performing FedAvg architecture. In the case of severe non-IID conditions, models gave up static identifiers. Learning was based on universal protocol dynamics such as `Rate`, `std_duration` and active temporal metrics.

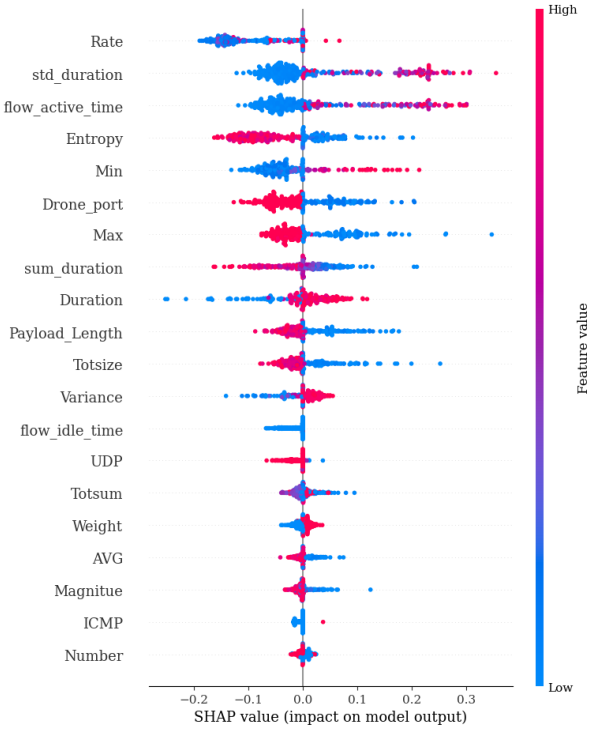

Fig. 6. SHAP beeswarm summary for the federated FedAvg model.

Fig. 7 shows federated aggregation acting as an implicit regularizer across heterogeneous domains. The global model prioritizes universal network parameters instead of memorizing trivial shortcuts.

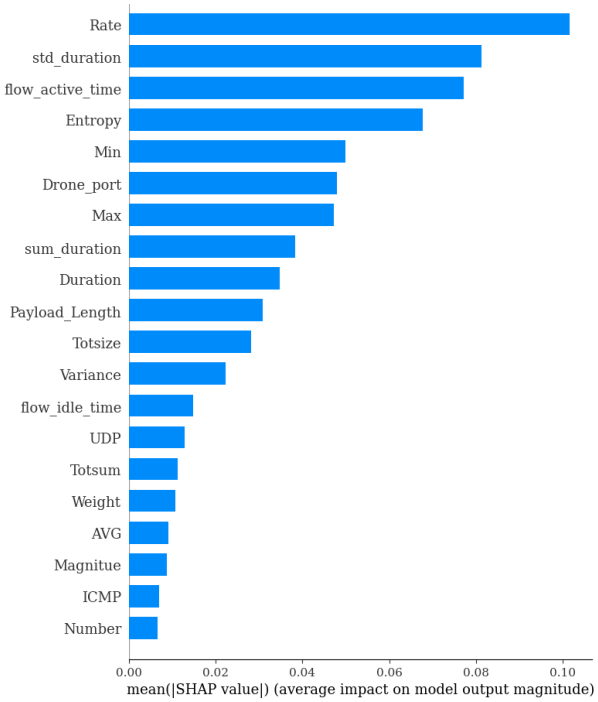

Fig. 7. SHAP global importance bar plot for federated FedAvg.

V. Conclusion and Future Work

Centralized models essentially run counter to strict swarm privacy restrictions. Federated learning provides an operational alternative that is much needed. FedAvg delivered the optimal score of 0.9844 macro. This is in full accord with the centralized performance ceiling. Severe feature skew was without effect to classification. As uncalibrated networks are nevertheless dangerous, in practice any calibration errors were removed by the applied temperature scaling. A 0.0026675 error rate permits one to safely go physically. Accurate probability distributions are directly in support of dynamic threat mitigation and Calibrated scores make it possible to have successful selective alerting mechanisms. A high confidence threshold is rigorously applied to hand ambiguous network traffic off to human analysts. Processing of clear threats autonomously de-dreads dreadful alarm fatigue. Explainability evaluations were used to prove decentralized architecture learns correctly. Centralized baselines memorized the ports and the federated models learn the universal solution anomalies in structure protocol. Reliance on basic flow metrics was corroborated by SHAP analysis. This behavior part is responsible for generalization to new network environments. Current Datasets are strictly determined by the structure of certain findings and Feature skew that is evaluated twisting a broad conclusions absolute. The adversarial client poisoning attacks need to be tested in future researches. Multiclass configurations should detect particular multiclass drone intrusion families. Adaptive algorithms will make dynamic aerial networks safe at all times.